

A Hierarchical Gene-Isoform Model for Sparse Single-Cell Data

Overview

The implemented method learns one low-dimensional state for each cell from paired equivalence-class (EC) counts. The same state controls both total gene abundance and conditional isoform usage. Primer-specific observation matrices then map the resulting transcript abundances to the observed ECs.

The fit has four stages:

1. initialize the cell state by log-normalized gene PCA;
2. fit the state and gene decoder to gene-unambiguous counts;
3. hold the state fixed while fitting the isoform decoder to paired EC counts;
4. jointly refine both decoders and the cell state while retaining the gene-level objective.

All parameters are shared genome-wide. The method does not estimate a free transcript vector in each cell and never stores a complete cell-by-transcript abundance matrix.

Observed Data

Let $m = 1, \dots, M$ index cells, $g = 1, \dots, G$ genes, and $t = 1, \dots, T$ transcripts. Let $g(t)$ be the gene containing transcript t , and let

$$\mathcal{T}_g = \{t : g(t) = g\}.$$

Primer type is indexed by p . The paired Parse Biosciences data have an oligo(dT) half and a random-hexamer half for each cell.

For primer p , the input consists of:

- n_{kmp} , the UMI count for EC k in cell m ;
- $N_{mp} = \sum_k n_{kmp}$, the observed UMI total;
- $A_p \in \mathbb{R}_+^{K_p \times T}$, a fixed EC-by-transcript observation matrix.

The nonzero entries of A_p combine transcript compatibility, Salmon conditional weights, positional-bias correction, and primer-specific sampling exposure. The oligo(dT) design uses no additional transcript-length factor. The random-hexamer design retains relative effective-length exposure because longer transcripts provide more priming opportunities.

For initialization and gene pretraining, define y_{gm} by summing counts from ECs whose compatible transcripts all belong to gene g , across both primer halves. Cross-gene ambiguous ECs are excluded from y_{gm} , but remain in the paired EC likelihood used to fit the isoform model.

Implemented Model

Shared cell state

Each cell has one latent state

$$\mathbf{z}_m \in \mathbb{R}^r.$$

There is no separate gene-expression state and splicing state. Two decoder heads map the same coordinates to gene abundance and isoform usage.

Gene abundance decoder

For gene g in cell m , define

$$\xi_{gm} = \alpha_g + \mathbf{u}_g' \mathbf{z}_m, \quad a_{gm} = \exp(\xi_{gm}).$$

Here α_g is a gene intercept and $\mathbf{u}_g$ is a row of the gene-loading matrix U . The positive quantity a_{gm} is the model's total abundance for gene g .

Within-gene isoform decoder

For transcript t , define

$$\eta_{tm} = \beta_t + \mathbf{w}_t' \mathbf{z}_m.$$

These logits are normalized only among transcripts from the same gene:

$$\pi_{tm} = \frac{\exp(\eta_{tm})}{\sum_{j \in \mathcal{T}_{g(t)}} \exp(\eta_{jm})}.$$

Thus

$$\sum_{t \in \mathcal{T}_g} \pi_{tm} = 1.$$

The transcript abundance is the product

$$\theta_{tm} = a_{g(t)m} \pi_{tm}.$$

Consequently, a_{gm} controls expression of the gene and π_{tm} controls allocation of that gene's abundance among its isoforms.

Expected EC mass

The expected unnormalized mass for EC k , cell m , and primer p is

$$q_{kmp} = \sum_t A_{pkt} \theta_{tm} = \sum_t A_{pkt} a_{g(t)m} \pi_{tm}.$$

This sum includes every compatible transcript, so within-gene and cross-gene ambiguous ECs need no separate handling in the main likelihood. The two primer halves share $\mathbf{z}_m$, a_{gm} , and π_{tm} , but use different observation matrices A_p .

Objectives

Gene objective

Let $Y_m = \sum_g y_{gm}$. Gene pretraining uses

$$\mathbf{y}_m \mid Y_m \sim \text{Multinomial} \left(Y_m, \frac{\mathbf{a}_m}{\sum_g a_{gm}} \right).$$

Ignoring constants, the mean negative log likelihood per included molecule is

$$\mathcal{L}_{\text{gene}} = \frac{-\sum_{m,g} y_{gm} \xi_{gm} + \sum_m Y_m \text{LSE}_g(\xi_{gm})}{\sum_m Y_m}.$$

Paired EC objective

Each primer half is conditioned on its own observed UMI total:

$$\mathbf{n}_{mp} \mid N_{mp} \sim \text{Multinomial} \left(N_{mp}, \frac{\mathbf{q}_{mp}}{\zeta_{mp}} \right), \quad \zeta_{mp} = \sum_k q_{kmp}.$$

The mean paired EC loss is

$$\mathcal{L}_{\text{EC}} = \frac{-\sum_{m,p} \sum_{k:n_{kmp}>0} n_{kmp} \log q_{kmp} + \sum_{m,p} N_{mp} \log \zeta_{mp}}{\sum_{m,p} N_{mp}}.$$

The implementation computes the normalizer without predicting every EC. Define the column-sum exposure

$$d_{pt} = \sum_k A_{pkt}.$$

Then

$$\zeta_{mp} = \sum_g a_{gm} \sum_{t \in T_g} d_{pt} \pi_{tm}.$$

For a column-normalized oligo(dT) design, $d_{pt} = 1$, and the normalizer reduces to $\sum_g a_{gm}$. The random-hexamer normalizer retains its isoform-dependent length exposure and is evaluated exactly.

Joint objective

The final stage minimizes

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{EC}} + \omega_g \mathcal{L}_{\text{gene}} + \frac{\lambda_w}{2} \text{mean}_{t,j}(w_{tj}^2).$$

Both likelihood terms are normalized per molecule. The implemented full run used $\omega_g = 1$ and $\lambda_w = 10^{-4}$. Retaining the gene term during joint refinement prevents transcript ambiguity from displacing marker-gene structure in the shared state.

Fitting Algorithm

Stage 1: sparse gene PCA

The gene-unambiguous count matrix $Y = (y_{gm})$ is normalized to 10,000 molecules per cell and transformed by $\log(1 + x)$. The 2,000 genes with largest variance are selected. Centered, variance-scaled randomized PCA produces an r -dimensional initial state Z . The calculation keeps the count matrix sparse and applies centering implicitly during matrix multiplication.

Stage 2: gene pretraining

The PCA state initializes z_m . Gene intercepts are initialized from pooled gene totals, and gene loadings start at zero. Adam then updates α , U , and Z using $\mathcal{L}_{\text{gene}}$.

Stage 3: fixed-state isoform fitting

The fitted Z , α , and U are held fixed. Isoform intercepts and loadings start at zero and are optimized using $\mathcal{L}_{\text{EC}}$. This stage learns the relationship between the existing cell state and conditional isoform usage without allowing EC ambiguity to move the cells.

Stage 4: joint refinement

All cell states and decoder parameters are unfrozen and optimized using $\mathcal{L}_{\text{joint}}$. The gene objective anchors expression structure while the paired EC objective can refine the state using isoform information.

The full-data implementation uses shuffled cell minibatches, Adam, and a gradient-norm limit of 10. The initial run used rank 32, 512 cells per minibatch, and ten epochs per stage. Learning rates were 10^{-2} for gene pretraining, 3×10^{-3} for fixed-state isoform fitting, and 10^{-3} for joint refinement.

Implementation Details

Sparse observed-event likelihood

For a minibatch of B cells, the decoder forms

$$\Xi = \alpha + UZ' \quad \text{and} \quad H = \beta + WZ'$$

with shapes $G \times B$ and $T \times B$. A segmented softmax over transcript-to-gene membership converts H to Π . A temporary $T \times B$ abundance block is then formed and released after the minibatch.

Only cell-EC pairs with nonzero observed counts are expanded through the sparse rows of A_p . Their transcript contributions are summed by indexed scatter operations. Unobserved EC predictions are unnecessary because their only contribution is represented by the column-sum normalizer above.

The implementation stores the $M \times r$ cell state explicitly but never stores $T \times M$ transcript abundances. One million cells with rank 32 require about 128 MB for a float32 state matrix. Decoder evaluation and sparse event expansion, rather than cell-state storage, dominate computation.

Identifiability

The conditional likelihoods have several invariances. The implementation uses the following conventions:

- center gene intercepts globally;
- center isoform intercepts within each gene;
- center every isoform-loading coordinate within each gene;
- center and standardize the initial columns of Z .

The within-gene softmax is an identifiability and interpretation device, not a global simplex constraint on transcript abundance. Latent rotations remain possible and do not alter fitted abundances if both decoder heads are rotated with Z .

Full-Data Assessment

The first full fit used 201,023 cells passing 500 UMIs in each primer half, 236,543 transcripts, and 54,227 genes. Gene loss decreased from 8.092 to 8.075. EC loss decreased from 13.631 to 13.336 during fixed-state isoform fitting and to 12.929 during joint refinement. Ten epochs for each stage completed in 23 minutes and used approximately 48 GB peak host memory.

Cell-type labels were used only after fitting. Evaluation used the same 157,006 labeled cells, donor-held-out folds, balanced logistic classifier, and silhouette calculation as the standard-analysis comparison:

Representation	Accuracy	Balanced	Macro F1	Silhouette
Standard gene pipeline	0.896	0.918	0.877	0.105
Hierarchical PCA initialization	0.893	0.911	0.873	0.109
Hierarchical gene fit	0.915	0.936	0.906	0.144
Hierarchical joint fit	0.915	0.936	0.906	0.146
Published scVI	0.953	0.973	0.953	0.192

The implemented hierarchical state exceeds the matched standard gene pipeline on these external metrics. The published scVI representation remains a stronger reference.

Model Selection and Downstream Analysis

Cell labels are external assessments and are not used to select rank, ω_g , or λ_w . These choices should be evaluated with held-out molecule counts partitioned before normalization. Useful held-out diagnostics include gene-unambiguous likelihood, within-gene EC likelihood, cross-gene ambiguous EC likelihood, and separate likelihoods for each primer. A rank path such as $r = 4, 8, 16, 32, 64$ can be fit from small to large, with a one-standard-error rule selecting the smallest competitive rank.

For differential isoform analysis, covariates enter the within-gene logits:

$$\eta_{tm} = \beta_t + \mathbf{w}'_t \mathbf{z}_m + \mathbf{x}'_m \boldsymbol{\gamma}_t.$$

The coefficients $\boldsymbol{\gamma}_t$ are centered within each gene. Donor replication must be represented by donor-level random effects or by donor-by-cell-type pseudobulk tests. Differential splicing remains a covariate effect on conditional isoform usage, not a second latent state.

Connections to Other Approaches

Flat transcript models

The decomposition

$$\theta_{tm} = a_{g(t)m}\pi_{tm}$$

would be only a reparameterization if every cell had unrestricted gene totals and isoform proportions. The implemented model is restricted because both quantities are generated from the same low-dimensional $\mathbf{z}_m$:

$$\log \theta_{tm} = \alpha_{g(t)} + \mathbf{u}'_{g(t)}\mathbf{z}_m + \beta_t + \mathbf{w}'_t\mathbf{z}_m - \text{LSE}_{j \in \mathcal{T}_{g(t)}}(\beta_j + \mathbf{w}'_j\mathbf{z}_m).$$

The final gene-specific log-sum-exp is nonlinear in $\mathbf{z}_m$, so this is not generally equivalent to a flat log-linear transcript decoder.

Cell-level EM

Independent cell-level EM would estimate a high-dimensional transcript vector from only hundreds or thousands of molecules. The hierarchical model instead estimates r cell coordinates while sharing all decoder parameters across cells. Cells with little isoform information remain near the pooled within-gene intercepts, while informative cells contribute to shared isoform-loading estimates.

Pooled or pseudobulk EM could still initialize β_t , but it is not part of the current implementation. The implemented isoform intercepts start at zero and are learned jointly across all cells.

Poisson observation models

The implemented multinomial likelihood conditions on each primer-specific cell total. An alternative is

$$n_{kmp} \sim \text{Poisson}(s_{mp}q_{kmp}),$$

where s_{mp} is a primer-by-cell exposure. This would allow total UMI counts to contribute to fitting but would require estimating or profiling the exposures. The conditional multinomial model avoids these nuisance parameters and matches the molecule-count folds used for model assessment.

Direct nonnegative factorization

Direct EC factorization uses a linear low-rank transcript representation and optimizes EC reconstruction without explicitly preserving gene totals. The hierarchical decoder inserts gene abundance as a modeled quantity and uses a gene likelihood during initialization and joint refinement. Subsequent flat-GLM ablations show that most of the original performance gap comes from normalized log-count initialization, variance scaling, and retaining a gene-count objective rather than from the hierarchical decomposition alone. The explicit gene and within-gene isoform heads give a smaller residual gain over that count-aware flat model.